Ken (Xiaokang) **Liu**

Software Engineer

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Summary

Software Engineer with 4+ years of experience - four years at AWS S3 on distributed storage, archival data, and large-scale data systems, now on General Motors' autonomous vehicle data platform. Experienced in designing high-throughput backend services and self-serve ML data platforms spanning ingestion, processing, and serving, with a record of measurable latency, reliability, and cost improvements, including a 75% reduction in p95 ingestion latency and $49M+ in CapEx savings.

Work Experience

Software Engineer *Seattle, WA*

General Motors · Autonomous Vehicle Data Platform *Aug 2026 - Present*

- Design the next gen distributed ML data platform (Ingestion and management) on GCP and open-source frameworks
- Partner with ML and data engineers to translate pain points into platform requirements, shipping self-serve capabilities
- Deliver technical projects owning major technical decisions and tradeoffs and contribute to the team's strategic roadmap
- Champion engineering and operational excellence across the platform, tightening design, testing, and on-call practices
- Reduced p95 data lake to data warehouse ingestion latency by **75% (8h to 2h)**, bringing the pipeline under its 4h SLA

Software Development Engineer *Seattle, WA*

Amazon Web Services · S3 *Feb 2022 - Jul 2026*

- Designed and built a **distributed data lifecycle management service** to clean up expired data copies used in checksum validation, improving storage efficiency and protecting upstream service profit margins
- Developed **scalable high-throughput backend APIs** for archival data deletion, eliminating orphaned data creation
- Built an automated orphaned data detection and deletion system, resulting in over **$49 million** in CapEx savings
- Re-architected the service request message processing component with integrated cloud database solutions, reducing host CPU usage and message processing time by **90% (from seconds to milliseconds)**
- Improved system performance and memory efficiency, reducing **30%** heap memory usage and saving **$760K** annually
- Upgraded the **Apache Spark** framework across EMR clusters, reducing runtime by **30%** and saving over **$130K** annually
- Established a cross-team rapid archival data restoration service, reducing **Glacier Flexible Retrieval** restore times by up to **85% (from hours to minutes)** without additional infrastructure cost to customers

Skills

Languages	Java, Kotlin, Python, C/C++, Scala, TypeScript, Ruby
Backend & Systems	Distributed Systems, REST APIs, Microservices, CI/CD, Asynchronous Processing
Data & Infrastructure	ML Data Platforms, Data Lakes & Warehouses, Apache Spark, Large-scale Data Processing
Cloud Platform	AWS (DynamoDB, EBS, EC2, ECS, EMR, Lambda, SQS) GCP (BigQuery, BigTable, Pub/Sub)
AI-assisted Engineering	Cursor, Codex, Claude Code, AWS Kiro, GitHub Copilot, DeepSeek Harness
Specialties	System Design, Performance Optimization, Scalability, Reliability, Technical Leadership

Publications

- Edgar Salazar, **Xiaokang Liu**, and Gonzalo Arce, "X-ray Compton backscattering imaging via structured light," Opt. Express 30, 15211-15226 (2022)
- Angela P. Cuadros, **Xiaokang Liu**, Paul E. Parsons, Xu Ma, and Gonzalo R. Arce, "Experimental demonstration and optimization of X-ray StaticCodeCT," Appl. Opt. 60, 9543-9552 (2021)

Education

University of Delaware *Newark, DE*

Master of Science, Electrical and Computer Engineering *Feb 2019 - Feb 2022*

Bachelor of Science, Electrical Engineering *Aug 2015 - Dec 2018*